

PRACTICAL NO :10

AIM:.Sensor Mote and Base Station Communication Study Develop a simple communication scenario between multiple sensor motes and a base station to study data aggregation and sink node operation.

WSL Ubuntu installed

```
sanjana@DESKTOP-FL4Q95A: /mnt/c/Users/IT-03
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\IT-03> wsl --shutdown
PS C:\Users\IT-03> wsl --update
Checking for updates.
A connection with the server could not be established
Error code: Wsl/UpdatePackage/WININET_E_CANNOT_CONNECT
PS C:\Users\IT-03> wsl --install -d Ubuntu-22.04
Downloading: Ubuntu 22.04 LTS
Installing: Ubuntu 22.04 LTS
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu-22.04'
Launching Ubuntu-22.04...
Provisioning the new WSL instance Ubuntu-22.04
This might take a while...
Create a default Unix user account: 03
Invalid username. A valid username must start with a lowercase letter or underscore, and can contain lowercase letters, digits, underscores, and dashes.
Create a default Unix user account: sanjana
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sanjana@DESKTOP-FL4Q95A:/mnt/c/Users/IT-03$
```

ping + ns Internet working, NS-2 (2.35) and NAM installed
nano wsn_ns2.tcl

```
IT-03@DESKTOP-FL4Q95A: ~
IT-03@DESKTOP-FL4Q95A:~$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=6.34 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=8.17 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 6.335/7.250/8.165/0.915 ms
IT-03@DESKTOP-FL4Q95A:~$ wsl --shutdown
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
IT-03@DESKTOP-FL4Q95A:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
115 packages can be upgraded. Run 'apt list --upgradable' to see them.
IT-03@DESKTOP-FL4Q95A:~$ sudo apt install ns2 nam
ns2 is already the newest version (2.35+dfsg-9build1).
nam is already the newest version (1.15-11).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 115
IT-03@DESKTOP-FL4Q95A:~$ ns
% exit
IT-03@DESKTOP-FL4Q95A:~$ nano wsn_ns2.tcl
IT-03@DESKTOP-FL4Q95A:~$ nano wsn_ns2.tcl
```

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```
GNU nano 8.7.1 wsn_ns2.tcl
set cbr($i) [new Application/Traffic/CBR]
$cbr($i) attach-agent $udp($i)
$cbr($i) set packetSize $pktSize
$cbr($i) set interval $interval
$ns at [expr 1.0 + 0.1*$i] "$cbr($i) start"
$ns at [expr $val(stop) - 1.0] "$cbr($i) stop"

#----- TRAFFIC: CH -> SINK -----
# agg = 0 : CH forwards 6 packets, agg = 1 : CH sends 1 aggregated packet
if {$agg == 0} { set nflows 6 } else { set nflows 1 }
for {set k 0} {$k < $nflows} {incr k} {
    set udpc($k) [new Agent/UDP]
    $ns attach-agent $node_0 $udpc($k)
    set null_s($k) [new Agent/Null]
    $ns attach-agent $node_7 $null_s($k)
    $ns connect $udpc($k) $null_s($k)
    set cbrc($k) [new Application/Traffic/CBR]
    $cbrc($k) attach-agent $udpc($k)
    $cbrc($k) set packetSize $pktSize
    $cbrc($k) set interval $interval
    $ns at [expr 1.5 + 0.1*$k] "$cbrc($k) start"
    $ns at [expr $val(stop) - 1.0] "$cbrc($k) stop"
}

#----- END -----
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    exit 0
}

for {set i 0} {$i < $val(nn)} {incr i} {
    $ns at [expr $val(stop) - 0.1] "puts \"Node $i residual energy = \${[ $node_($i) energy-model] } energy\" \\\n"
}
$ns at $val(stop) "$ns nam-end-wireless $val(stop)"
$ns at [expr $val(stop) + 0.1] "finish"
puts "Starting simulation, aggregation = $agg"
$ns run
```

ns wsn_ns2.tcl 0

```
it-03@DESKTOP-FL4Q95A:~$ sed -i '/residual energy/d' wsn_ns2.tcl
it-03@DESKTOP-FL4Q95A:~$ ns wsn_ns2.tcl 0
num_nodes is set 8
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Starting simulation, aggregation = 0
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
```

ns wsn_ns2.tcl 1

```
it-03@DESKTOP-FL4Q95A:~$ ns wsn_ns2.tcl 1
num_nodes is set 8
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Starting simulation, aggregation = 1
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
```

Packets received at sink: 586 and 98

```
it-03@DESKTOP-FL4Q95A:~$ awk '$1=="r" && $3=="_7_" && $4=="AGT"' wsn_agg0.tr | wc -l
586
it-03@DESKTOP-FL4Q95A:~$ awk '$1=="r" && $3=="_7_" && $4=="AGT"' wsn_agg1.tr | wc -l
98
```

Energy consumed per node, without aggregation

```
it-03@DESKTOP-FL4Q95A:~$ awk '{for(i=1;i<=NF;i++) if($i=="energy") e[$3]=$i+1} END{for(n in e) printf "%s consumed = %.4f \\\n", n, 5-e[n]}' wsn_agg0.tr | sort
0_ consumed = 1.7667 J
1_ consumed = 1.7676 J
2_ consumed = 1.7689 J
3_ consumed = 1.7701 J
4_ consumed = 1.7714 J
5_ consumed = 1.7733 J
6_ consumed = 2.2623 J
7_ consumed = 1.7972 J
```

Energy consumed per node, with aggregation

```
it-03@DESKTOP-FL4Q95A:~$ awk '{for(i=1;i<=NF;i++) if($i=="energy") e[$3]=$i+1} END{for(n in e) printf "%s consumed = %.4f \\\n", n, 5-e[n]}' wsn_agg1.tr | sort
0_ consumed = 1.0786 J
1_ consumed = 1.0802 J
2_ consumed = 1.0816 J
3_ consumed = 1.0836 J
4_ consumed = 1.0847 J
5_ consumed = 1.0880 J
6_ consumed = 1.1845 J
7_ consumed = 1.0244 J
```